1. Clinical Question & Core Hypothesis

Clinical Question

What molecular mechanisms determine whether immune structures translate into clinical benefit?

Working Hypothesis

Expanded hypothesis: There exists a human bladder cancer biomarker network in which a prioritized set of nodes—MKI67, TP53, RB1, FGFR3, CCND1, CDKN2A, ERCC2 and related DNA-damage repair factors, PD-L1/CD274 and other immune-checkpoint genes, EMT markers (CDH1, SNAI1, ZEB1), and APOBEC-signature–related nodes—exhibits distinct dysregulation patterns between NMIBC and MIBC. The network will be characterized by a combination of direct mechanistic edges (e.g., regulatory interactions, mutation-driven pathway activation) and strong co-expression/prognostic edges (e.g., edges that correlate with recurrence, progression, chemotherapy or immunotherapy response). Specific testable predictions include: (1) differential expression and mutation spectra of the gene set between NMIBC and MIBC that associate with recurrence-free and progression-free survival; (2) DDR/TP53/RB1 alterations and FGFR3 status predicting chemo-sensitivity (cisplatin-based regimens) and, separately, responsiveness to immune-checkpoint blockade; (3) immune-checkpoint and EMT signatures correlating with tumor immune microenvironment status (inflamed vs. excluded) and predicting response to BCG or PD-1/PD-L1 inhibitors; (4) integration of mechanistic and co-expression edges into a network whose edge weights stratify patients by prognosis and therapy response, enabling a concise biomarker model capped at ~30 genes/edges. Edge metadata will be captured (evidence type: literature, curated databases, and patient cohorts) to enable Tier 1 high-confidence edges. Experimental testing plan: (i) assemble human bladder cancer cohorts with explicit NMIBC/MIBC annotation and comprehensive omics data (transcriptomics, mutation, CNV) plus clinical outcomes and treatment history; (ii) perform differential expression/mutation analyses for the prioritized genes and compute associations with recurrence/progression and therapy response; (iii) estimate immune infiltration and examine co-expression patterns with immune/EMT markers; (iv) construct a multi-layer network integrating curated regulatory/pathway edges and data-driven co-expression edges, weight edges by supporting evidence, and derive a predictive signature using penalized regression or network-based learning; (v) validate the top edges and the predictive model in independent cohorts and, where feasible, corroborate with protein-level assays (IHC, targeted proteomics).

2. Why This Gene — Not Others

3. Clinically Testable Predictions

Prediction 1: , cell-cycle regulators (TP53, RB1, CCND1, CDKN2A), FGFR3 mutations, and DNA-damage repair genes (ERCC2, other DDR factors)—show distinct patterns across NMIBC and MIBC and influence treatment outcomes, including cisplatin-based chemotherapy. Immune checkpoint genes (e.g., PD-L1/CD

Prediction 2: and EMT markers (CDH1, SNAI1, ZEB

Prediction 3: shape the tumor immune microenvironment and have been linked to responses to BCG and PD-1/PD-L1 inhibitors. APOBEC mutagenesis is a prominent feature in bladder cancer and associates with mutational burden and prognosis.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions